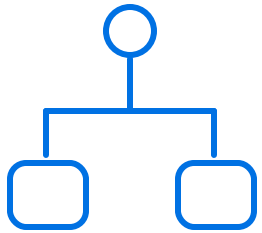
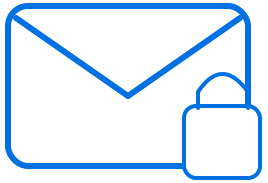
A fairer comparison.



Stratified randomisation

Helps balance sex and BMI across the two groups.



Concealed allocation

Reduces enrolment choices based on the next assignment.



Shared support



Calibrated weighing

A clearer signal.



Shared support

Makes unequal attention
a less likely explanation.



Calibrated weighing

Makes weight less dependent
on participants' self-report.



Stratified randomisation



Concealed allocation

Lim et al. 2024 · Methods 2.4–2.7 · Each feature strengthens a different part of the comparison.

More weight loss.

But why?

The mechanism was not isolated.

Lim et al. 2024 · Table 2, Table 5 and Discussion · Programme effect and mechanism are different questions.

Two diets. Different reductions.

HKD

−704

kcal/day

ERD

−423

kcal/day

Reported change from baseline at 6 months

Both advised energy restriction. Actual intakes were not matched.

Lim et al. 2024 · Table 5 · Self-reported intake changes, not assigned or absolute calorie intakes.

Assigned does not mean verified.



Diet advice and food diaries do not confirm sustained ketosis.
Dashed arrows = a possible pathway.

Lim et al. 2024 · Methods 2.3, 2.7; Table 5; Discussion · No ketone verification reported.

Two possible routes. One result.



Possible paths can overlap. Their contributions were not isolated.
Dashed arrows = hypotheses, not demonstrated mediation.

Lim et al. 2024 · Methods 2.3, 2.7; Table 5; Discussion · No ketone verification reported.

A programme effect.

An unresolved mechanism.

Ketosis unverified

Was the state achieved?

Calories unmatched

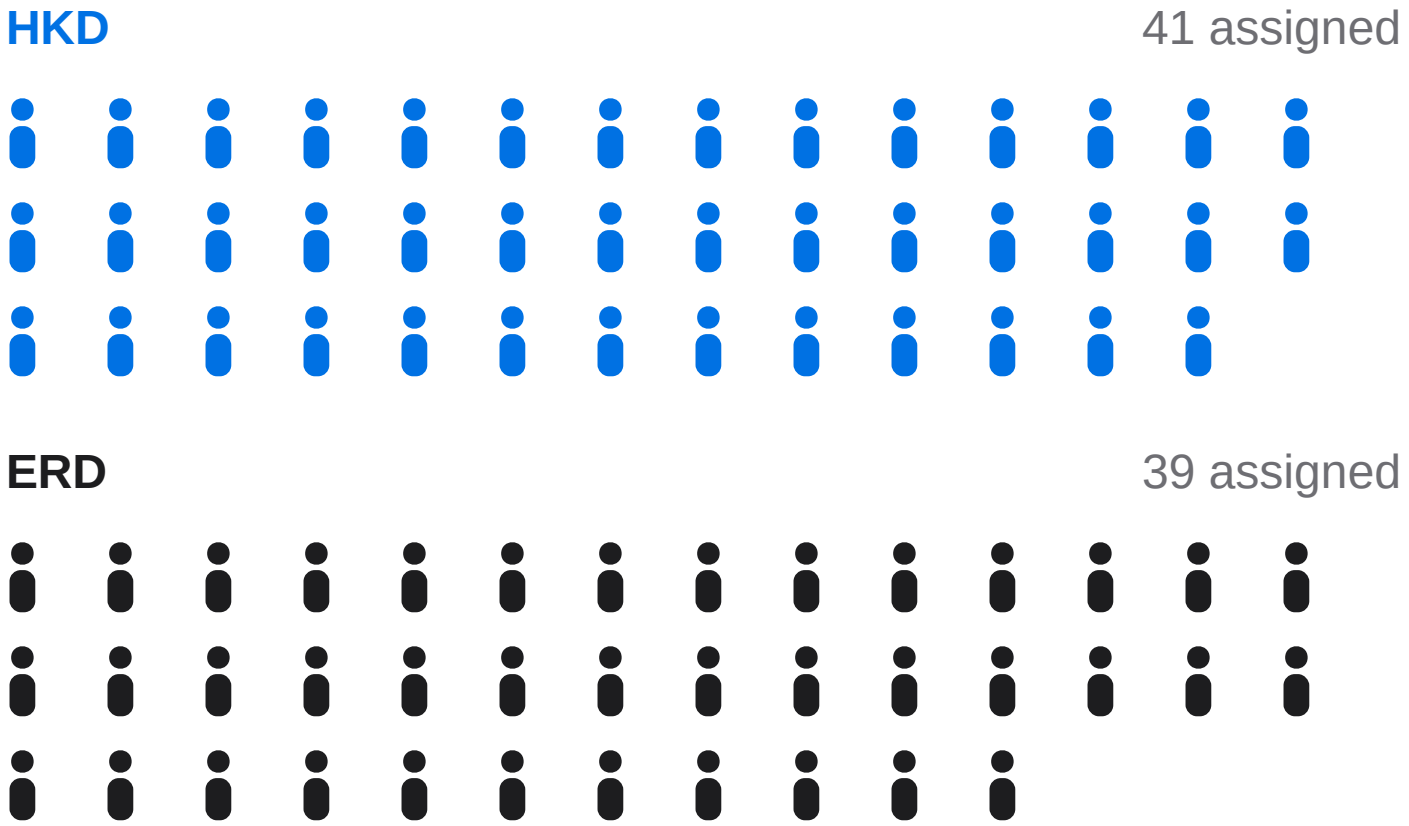
Would extra loss persist?

Eating less may be part of the programme’s benefit.

A calorie-independent ketosis benefit was not established. The programme comparison can still be useful.

80

people randomised



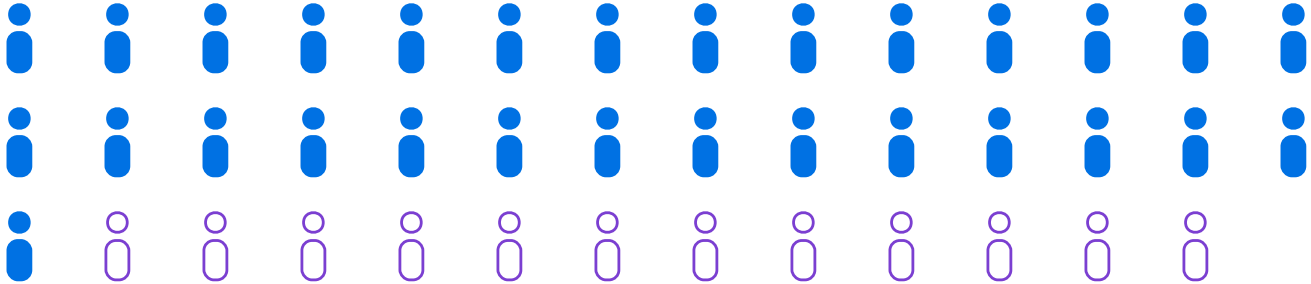
The same participants will remain in view.

Lim et al. 2024 · Figure 1 and Table 2 · Each participant symbol represents one person.

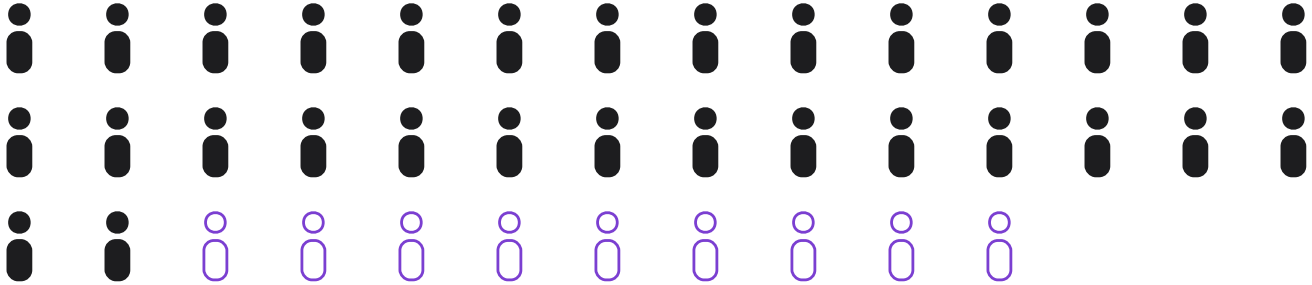
59

weights available at 6 months

HKD 29 / 41 measured



ERD 30 / 39 measured



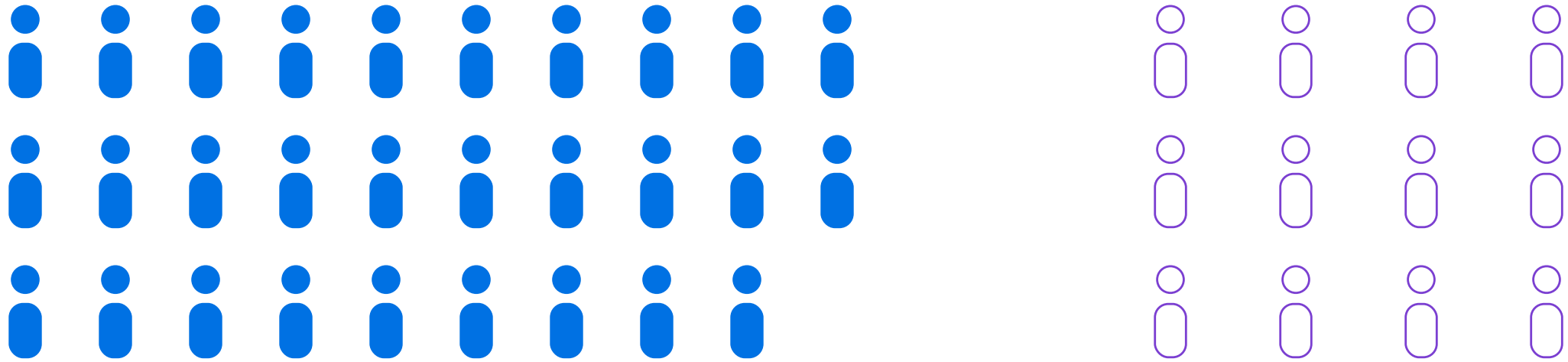
21 missing outcomes. Missing measurement ≠ not following the diet.

Lim et al. 2024 · Figure 1 and Table 2 · Each participant symbol represents one person.

Zoom in on HKD.

41 assigned

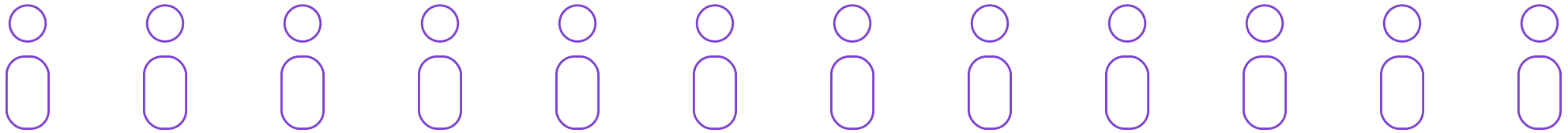
29 measured + 12 unknown



They might have followed the diet. Their weight outcomes are missing.

Lim et al. 2024 · Figure 1 · The 12 missing HKD outcomes are not automatically 12 diet failures.

Their weight loss is unknown.



12 unobserved HKD outcomes

Less?	Similar?	More?
weight loss	weight loss	weight loss

Their unknown outcomes may change the estimated benefit.

Qualitative possibilities, not observed results or invented participant-level weight values.

Who is missing matters.

IF poorer HKD responders were disproportionately missing…



Less successful outcomes absent → **Observed subset looks more successful** → **Benefit may be overstated**

Hypothetical condition

Potential attrition bias — this is a conditional example, not a finding.

Illustration for a comparison based on measured participants; actual missing outcomes and bias direction are unknown.

Smaller benefit?

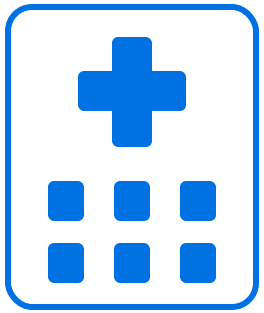


Larger benefit?

The direction is unknown.

Missing-outcome handling was not clearly explained.

One setting. A selected sample.



One hospital
Healthcare staff

Singapore

86%

women

69 of 80 participants



English literacy
Smartphone access

Required to participate

Random assignment ≠ a representative population sample.

Lim et al. 2024 · Methods 2.2 and Table 1 · Some diseases were also excluded.

Support shaped the setting.

The same support structure in both groups

The intervention was delivered under these conditions.

7

workshops



App + dietitian coaching

6

months of support

The result belongs to this programme and its delivery conditions.

Same benefit elsewhere?

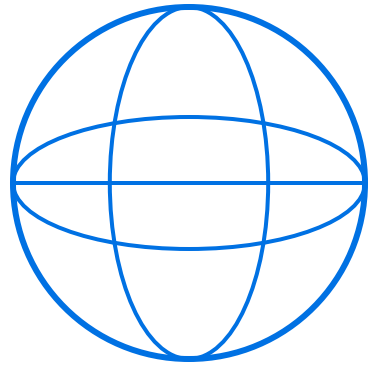
Selected people

One hospital

7 workshops

App + dietitian support





Other adults?

Less support?

Applicability is untested. That does not mean no benefit.

The dashed bridge asks about transfer across people and settings; it is not a biological pathway.

Three limits. Three questions.



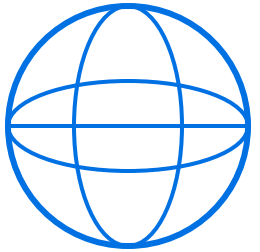
Why?

Mechanism



How reliable?

Missing outcomes



For whom?

Generalisability

A useful programme comparison. Carefully bounded conclusions.

Lim SL et al. (2024), Nutrients 16:4380 · doi:10.3390/nu16244380